

deepened_round2

Deepened inventory (round 2)

Round-2 deepening of packages flagged shallow/partial/unknown in round 1 (internal_services.md coverage notes + the owned-submodule umbrella rows). Each previously-thin entry is enumerated into discrete features in the SAME 12-column schema as internal_services.md (legend in docs/features/Status.md § “Status dimensions”). Assessed from real code evidence (impl reality, wiring, *_test.go presence) per CONST-035 / §11.4.107 anti-bluff.

Anti-bluff caveats (read before trusting a row). - 📺 Video is no for every row — no analyzed recording exists; the conductor owns video confirmation. Overall is therefore **never** confirmed; honest rollups are working-untaped / partial / gap. - Origin is native (all own-org code). - **Round-1 stub claims corrected here from direct source reads** (a round-2 exploration subagent reported these as stubs WITHOUT reading the code — verified false): workspace docker Stop/Remove are real (docker stop / docker rm -f, manager.go:21-37); voice recorder Stop is real (Process.Signal(os.Interrupt) then Kill(), recorder.go:63-80); roocode CodeReviewer.Review is real (reads the file + scans for TODO/issue patterns, reviewer.go:16-30). They are done/partial (impl real) but Real-use=unknown (shipped-flow wiring unconfirmed), not stubs. - Submodule-side Wired follows internal_services.md’s wiring model: helix_agent is imported directly (Wired=yes where HelixCode consumes it), transitively-reachable own-org deps are Wired=partial, repo-present-but-not- in-build-graph are Wired=no. helix_specifier/helix_qa are imported by module path (Wired=yes); panoptic is a Challenge/recording submodule not in HelixCode’s Go build graph (Wired=no). - Submodule Tests reflects *_test.go presence in/near the package; it is NOT a coverage-percentage claim. README-cited coverage figures (helix_agent “65.6%”, panoptic “78%”) are NOT used as evidence — unverified.

Internal packages (round-1 partial/unknown → deepened)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
infrastructure	internal/clientcore	agentic tool registry wiring (git/fs/LSP/MCP, live-built)	done	yes	yes	unit	no
infrastructure	internal/clientcore	LSP read-only diagnostics tool wiring	done	yes	unknown	unit	no
infrastructure	internal/clientcore	MCP config-merge + server startup + tool registration	done	yes	unknown	unit	no
infrastructure	internal/clientcore	tool-loop system-prompt generation (from registry names)	done	yes	yes	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
infrastructure	internal/ clientcore	tool-trace adapter (agent → ensembleui)	done	yes	yes	unit	no
infrastructure	internal/ clientcore	skills + plugins loader (graceful-on-failure)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	verifier adapter wiring (CONST-036/037 single-source)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	env-provider registration (cloud API keys, zero-hardcode)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	HelixAgent provider registration (when server reachable)	done	partial	unknown	unit	no
infrastructure	internal/ agentbridge	verifier bridge config wrapper	done	yes	unknown	unit	no
infrastructure	internal/ agentbridge	VerifyModel real HTTP call to verifier service	done	yes	unknown	unit	no
infrastructure	internal/ agentbridge	test request/result marshal+unmarshal	done	yes	unknown	unit	no
infrastructure	internal/ checkpoint	git-backed snapshot (stash/write-tree/read- tree)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	file-copy fallback backend (real os.Read/ Write)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	checkpoint create (capture)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	checkpoint restore (undo, writes real bytes)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	metadata persistence (label/timestamp JSON)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	multi-backend auto- selection (git else files)	done	yes	yes	unit	no
service	internal/ ensembleui	tool-trace formatting (text/markdown render)	done	yes	unknown	unit	no
service	internal/ ensembleui	metadata extraction helpers (int/string/slice)	done	yes	unknown	unit	no
service	internal/ ensembleui	multi-model response assembly	done	yes	unknown	unit	no
service	internal/ ensembleui	per-model token-usage aggregation	done	yes	unknown	unit	no
infrastructure			done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
	internal/ substrate	Unit interface (task contract)					
infrastructure	internal/ substrate	queue-based scheduler (enqueue + dispatch)	done	yes	unknown	unit	no
infrastructure	internal/ substrate	serial FIFO execution (single worker)	done	yes	unknown	unit	no
infrastructure	internal/ substrate	context-cancellation propagation to Execute	done	yes	unknown	unit	no
infrastructure	internal/ workspace	workspace create (project bootstrap)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	docker runner start/stop/remove/list (real os/exec)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	workspace status tracking (enum + DB persist)	done	partial	unknown	unit	no
infrastructure	internal/ workspace	CLI tools (create/delete/list registration)	done	yes	unknown	unit	no
infrastructure	internal/ workspace	il8n translator seam	done	yes	yes	unit	no
infrastructure	internal/ workspace	auto-cleanup / TTL teardown	partial	no	unknown	unit	no
service	internal/ voice	audio device detection (arecord/sox/parec probe)	done	partial	unknown	unit	no
service	internal/ voice	audio capture start (real exec.Command)	done	partial	unknown	unit	no
service	internal/ voice	audio capture stop (Signal(Interrupt)→Kill)	done	partial	unknown	unit	no
service	internal/ voice	Whisper API transcription (real multipart HTTP POST)	done	partial	unknown	unit	no
service	internal/ voice	WAV validation before transcribe	done	partial	unknown	unit	no
service	internal/ voice	local whisper.cpp fallback transcription	partial	no	unknown	unit	no
service	internal/ roocode	in-memory conversation store (create/add-message)	done	partial	unknown	unit	no
service	internal/ roocode	task delegation (TaskSpec build; no real dispatch)	partial	no	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
service	internal/ roocode	code generation (Generate/Bootstrap scaffold)	partial	no	unknown	unit	no
service	internal/ roocode	code review (file read + TODO/issue scan)	done	partial	unknown	unit	no
service	internal/ roocode	il8n translator seam	done	yes	yes	unit	no
service	internal/ telemetry	OpenTelemetry provider bootstrap (OTLP exporter)	done	yes	unknown	unit	no
service	internal/ telemetry	agent instrumentation (spans + iteration counter)	done	yes	unknown	unit	no
service	internal/ telemetry	LLM instrumentation (traced-provider wrapper)	done	yes	unknown	unit	no
service	internal/ telemetry	tool instrumentation (call spans)	done	yes	unknown	unit	no
service	internal/ telemetry	config-from-env (OTEL_* vars + exporter-kind resolve)	done	yes	yes	unit	no
infrastructure	internal/ verifier	flexible timestamp parse (Unix + RFC3339)	done	yes	yes	unit	no
infrastructure	internal/ verifier	adapter facade (cache+health+client, IsEnabled/Reachable)	done	yes	unknown	unit	no
infrastructure	internal/ verifier	two-tier cache (TTL + Redis)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	health monitor (background liveness goroutine)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	background poller (interval model-sync)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	bootstrap (embedded-or- remote stack init)	done	yes	unknown	unit,integ	no
infrastructure	internal/ verifier	embedded verifier server (fallback)	partial	partial	unknown	unit	no
infrastructure	internal/ verifier	fallback model list (offline mode)	done	yes	yes	unit	no
infrastructure	internal/ verifier	real-server integration test (live LLMsVerifier)	done	yes	unknown	integ	no
service	internal/ worker		done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		consensus manager (Raft state machine, election/heartbeat)					
service	internal/worker	vote-transport abstraction (decoupled from wire)	done	partial	unknown	unit	no
service	internal/worker	single-node fallback (nil-transport safe step-down)	done	yes	yes	unit	no
service	internal/worker	worker isolation: sandbox dir create (0750)	done	partial	unknown	unit	no
service	internal/worker	worker isolation: resource limits (cgroup writes)	done	partial	unknown	unit	no
service	internal/worker	SSH client pool (persistent connection reuse)	done	yes	yes	unit	no
service	internal/worker	consensus stress + chaos tests (concurrent votes, SIGKILL)	done	yes	yes	unit	no
service	internal/worker	worker-pool isolation stress test	done	yes	yes	unit	no
service	internal/server	Gin server + JWT auth middleware	done	yes	unknown	unit,integ	no
service	internal/server	project list/register handlers (user-from-context)	done	yes	yes	integ	no
service	internal/server	LLM generate endpoint (real provider stream)	done	yes	yes	unit,integ	no
service	internal/server	LLM model-list endpoint (verifier-queried)	done	yes	yes	unit,integ	no
service	internal/server	worker/task/project CRUD handlers	done	yes	yes	integ	no
service	internal/server	QA session start handler (helix_qa integration)	done	yes	unknown	unit	no
service	internal/server	specify endpoint (spec from prompt)	done	yes	yes	unit,integ	no
service	internal/server	stats / pprof / error-categorize (400-vs-500) helpers	done	yes	yes	unit	no
service			done	yes	yes	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	
	internal/ server	i18n translator seam (context-aware)						

Owned submodules (round-1 umbrella rows → deepened principal packages)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	
submodule	helix_agent	internal/llm multi-provider interface (Claude/DeepSeek/Gemini/Mistral/Qwen/xAI/OpenRouter/Ollama/llama.cpp/Bedrock/Azure)	done	yes	unknown	unit,integ	no	no
submodule	helix_agent	internal/ensemble multi-model debate/fusion orchestrator	done	yes	yes	unit,integ,e2e	no	no
submodule	helix_agent	internal/provider+providers per-provider abstraction layer	done	yes	unknown	unit	no	no
submodule	helix_agent	internal/verifier LLMsVerifier single-source-of-truth	done	yes	unknown	unit	no	no
submodule	helix_agent	internal/server HTTP(Gin) +WebSocket API (completions/ensemble/stream)	done	yes	unknown	unit,integ,e2e	no	no
submodule	helix_agent	internal/handlers per-route request/response handlers	done	yes	unknown	unit	no	no
submodule	helix_agent	internal/database PostgreSQL persistence (pgx/v5, migrations)	done	partial	unknown	integ	no	no
submodule	helix_agent	internal/redis caching layer (go-redis/v9)	done	partial	unknown	integ	no	no
submodule	helix_agent	internal/auth JWT + bcrypt/argon2 + OAuth2	done	partial	unknown	unit,integ	no	no
submodule	helix_agent		done	partial	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	🔒 V
		internal/security PII redaction + guardrails bridge						
submodule	helix_agent	internal/streaming SSE/WebSocket token-level streaming	done	yes	unknown	unit	no	no
submodule	helix_agent	internal/cache semantic (embeddings dedup) caching	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/knowledge RAG / vector-DB (zep) bridge	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/ monitoring Prometheus metrics	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/logging structured JSON + gRPC tracing	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/mcp Model Context Protocol server/ client	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/ skills+plugins hot- reload plugin system	done	partial	unknown	unit	no	no
submodule	helix_agent	internal/analytics usage telemetry + warehouse export	partial	partial	unknown	unit	no	no
submodule	helix_agent	pkg/sdk Go + Python client SDKs	done	partial	unknown	unit	no	no
submodule	security	pkg/guardrails content guardrail engine (severity rules)	done	partial	unknown	unit	no	no
submodule	security	pkg/pii PII detect+redact (email/phone/SSN/ CC-Luhn/IPv4)	done	partial	unknown	unit	no	no
submodule	security	pkg/content composable filter	done	partial	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	
		chains (length/ pattern/keyword)						
submodule	security	pkg/policy rule- based policy enforce (allow/ deny/audit)	done	partial	unknown	unit	no	no
submodule	security	pkg/scanner vuln- scan interface + report aggregation	done	partial	unknown	unit	no	no
submodule	security	pkg/headers HTTP security headers middleware (CSP/ HSTS/...)	done	partial	unknown	unit	no	no
submodule	security	pkg/securestorage AES-256-GCM + Argon2id key derivation	done	partial	unknown	unit	no	no
submodule	security	pkg/e2ee end-to- end encryption (ChaCha20- Poly1305)	done	partial	unknown	unit	no	no
submodule	security	pkg/ssrf SSRF prevention (outbound HTTP guard)	done	partial	unknown	unit	no	no
submodule	security	pkg/attestation + pkg/gpuattest platform/GPU attestation	partial	no	unknown	unit	no	no
submodule	helix_specifier	parallel execution (bounded- concurrency dispatch)	done	yes	unknown	unit	no	no
submodule	helix_specifier	constitution-as- code (machine- readable rule enforce)	done	yes	unknown	unit	no	no
submodule	helix_specifier	Nyquist TDD ($\geq 2\times$ test-to-impl ratio enforce)	done	yes	unknown	unit	no	no
submodule	helix_specifier	debate architecture (multi-round spec refinement)	done	yes	yes	unit	no	no
submodule	helix_specifier	skill learning (running-avg	done	yes	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	
		proficiency tracking)						
submodule	helix_specifier	brownfield analysis (legacy pattern/dep detection)	done	yes	unknown	unit	no	no
submodule	helix_specifier	predictive specification (future-req prediction)	done	yes	unknown	unit	no	no
submodule	helix_specifier	cross-project transfer (shared knowledge base)	done	yes	unknown	unit	no	no
submodule	helix_specifier	adaptive ceremony (dynamic level by quality metrics)	done	yes	unknown	unit	no	no
submodule	helix_specifier	spec memory (persistent index + semantic search)	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/autonomous autonomous QA session engine (~9.5k loc, 21 tests)	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/orchestrator QA run orchestration	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/navigator UI navigation/driving (~3.8k loc, 16 tests)	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/detector + pkg/issuedetector issue/error detection	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/recordingqa recording-validator (anti-bluff §11.4.107)	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/regression standing regression-guard suite	done	yes	unknown	unit	no	no
submodule	helix_qa		done	yes	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
		pkg/evidence captured-evidence management						
submodule	helix_qa	pkg/challengegen Challenge generation	partial	yes	unknown	unit	no	no
submodule	helix_qa	pkg/reporter QA reporting	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/conduit + pkg/ bridge sync- channel / agent bridge (§11.4.116)	done	yes	unknown	unit	no	no
submodule	helix_qa	pkg/replay + pkg/ reproduce replay / defect reproduction	done	partial	unknown	unit	no	no
submodule	helix_qa	pkg/distributed + pkg/maestro distributed/maestro coordination	partial	partial	unknown	unit	no	no
submodule	helix_qa	banks/ test-bank corpus (JSON/ YAML suites)	done	yes	unknown	none	no	no
submodule	helix_qa	cmd/* tooling (axtree/capture/ recvalidate/ omniparser/uitars/ lpips/...)	done	partial	unknown	unit	no	no
submodule	panoptic	internal/launcher multi-platform test launcher (web/ desktop/mobile)	done	no	unknown	unit	no	no
submodule	panoptic	internal/executor UI automation executor	done	no	unknown	unit	no	no
submodule	panoptic	internal/vision computer-vision element detection	done	no	unknown	unit	no	no
submodule	panoptic	internal/ocr OCR text extraction	done	no	unknown	unit	no	no
submodule	panoptic	internal/recvalidate recording validator	done	no	unknown	unit	no	no
submodule	panoptic	internal/ai AI test generation/ enhancement	done	no	unknown	unit	no	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V	V
submodule	panoptic	internal/cloud cloud storage/ integration	done	no	unknown	unit	no	no
submodule	panoptic	internal/platforms per-platform automation adapters	done	no	unknown	unit	no	no
submodule	panoptic	screenshot + video recording capture	done	no	unknown	unit	no	no
submodule	panoptic	cmd/record + cmd/ vision + cmd/ testgen CLI tooling	done	no	unknown	unit	no	no

Round-2 coverage notes

- **Internal deepening:** the 12 round-1-flagged packages (clientcore, agentbridge, checkpoint, ensembleui, substrate, workspace, voice, roocode, telemetry, verifier, worker, server) are now broken out into 73 discrete feature rows. checkpoint, telemetry, most of verifier, server, and several worker features upgrade from round-1 partial to working - untaped on direct-source evidence (real git plumbing, real OTEL exporter, real Raft + cgroup writes, real Gin handlers with integration tests). roocode task-delegation/code-gen and voice/workspace auto-cleanup remain honest gap (real types, no shipped wiring / incomplete impl).
- **Stub-claim corrections** (see top caveat): three round-2-explorer “stub” reports were verified FALSE by direct read and corrected to done/partial. Logged here so the conductor does not propagate the wrong call.
- **Submodule Real-use=unknown throughout:** each submodule package compiles and has tests, but genuine end-user reachability THROUGH HelixCode (vs the submodule’s own standalone tests) was not confirmed by static inspection — honestly unknown, never green.
- **helix_agent Wired:** yes for the packages HelixCode consumes via the direct dev.helix.agent import (llm/ensemble/server/provider/verifier/ streaming surface reachable through the agent dependency); partial for packages reachable only transitively (database/redis/auth/monitoring/etc).
- **panoptic Wired=no:** present as an equal-codebase submodule (CONST-051) but NOT in HelixCode’s Go build graph — it is a Challenge/recording framework consumed out-of-process, not imported.
- Every Real-use=unknown / working - untaped row is a candidate for a recorded scenario; none is video-confirmed (📹 no throughout).

Round-2 deepened feature count: 145 rows (73 internal across 12 packages + 72 submodule across helix_agent 19, security 11, helix_specifier 10, helix_qa 14, panoptic 10, distributed across the wiring model).